

Fraudulent Review Identification Using Graph Embeddings and Metaheuristic Optimization Techniques

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Abstract—Online reviews play crucial role in spreading opinions, guiding purchases, and impacting a company’s reputation. However, these reviews are sometimes doubted because some fake ones end up misleading consumers. This paper presents a hybrid approach for fake review detection by combining semi-supervised and supervised learning techniques to enhance classification performance. The proposed approach utilizes GraphSAGE and Graph autoencoders to create a strong framework capable of learning patterns in review connections, creating embeddings that help Whale optimization algorithm (WOA) and LightGBM classifier to classify reviews. The preprocessing includes cleaning, lemmatization, stop-word removal, and stemming. Then feature extraction enriches this by including sentiments analysis, Bidirectional Encoder Representations from Transformers (BERT), Principal component analysis (PCA) scores. The study aims to address the limitations of traditional supervised methods, which require ample labelled datasets and often struggle with scalability. The model’s parameters, that were optimized using Whale optimization algorithm, achieved a high accuracy of 98%. Experimental results show that our hybrid approach achieved better classification accuracy and efficiency compared to other state-of-the-art methods, neural network architectures like graph neural networks, even with unlabeled data.

Index Terms—Fake reviews, Graph Autoencoder, GraphSAGE, Natural language processing, Whale optimization

I. INTRODUCTION

The online e-commerce platform has been a force to transform consumer-decision making on purchases, and a key contributory factor to these choices is online reviews from other consumers pertaining to the products. These reviews thus represent some of the most valuable sources of information on product quality and reliability. It has also led to the more dangerous fraudulent reviews that aim to cheat people into making choices for personal or corporate interests. Those fake reviews may be employed to artificially improve the reputation of a product or blemish the reputation of a competitor thus offering a serious challenge to the integrity of online markets. The fraudulent reviews impact are widespread. Positive fraudulent reviews may inflate the attractiveness of a product, and then its sales may skyrocket, and revenue for merchants tends to climb up as well. On the other hand, fake negative reviews can damage the reputation of the competitor, and that increase some financial loss besides it. Not only this fraudulent activity misleads consumers but also destroys their trust in these online platforms. Several existing studies have not been

able to, in a first attempt, be able to deeply probe into the motivations behind fake positive reviews and negative reviews. Instead, they have failed to effectively integrate multi-modal features-such as textual data and sentiment-into their detection models. In this respect, fraudsters find it more challenging using increasingly sophisticated methods. In such a case, there is an increased need for better fraud detection mechanisms to detect fake reviews in real time and protect consumers and businesses from such practices.

We apply GraphSAGE - graph-based model in our study to solve the above issues by detection of fraudulent patterns on labeled reviews datasets. In a step describing the dataset as a graph, we use each review as a node such that nodes are connected based on semantic similarity of their reviews. GraphSAGE generates embeddings by sampling and aggregating information from neighboring nodes with both local context and relational information between reviews. This model is specifically suitable for applying in large-scale labeled datasets without computing all the possible connections within the graph but sampling and aggregating a subset of nearby nodes. Labeled dataset where GraphSAGE picks up review interactions to finally classify fraud and authentic reviews.

GraphSAGE performs well on labeled datasets but the real task is dealing with the unsupervised set of datasets, where reviews have no explicit labels about being fraudulent or not. In this regard, we use a graph autoencoder for unsupervised learning. With this approach, in each review, a node is represented with edges regarding content similarity, and the goal of the model here is to learn the latent representations of the graph structure itself. The approach here compresses the graph into a lower dimensional space by learning embeddings for each of its nodes and then reconstructing the graph from these embeddings. This allows the model to identify novel patterns and detect anomalous reviews, even without explicitly being labeled as fraud. Once we have embedded the reviews, we assign pseudo-labels based on detected anomalies, and then we can combine these pseudo-labeled reviews with our labeled data to fine-tune the detection model. Figure 1 below visualizes the process of how a graph autoencoder identifies anomalous reviews in an unlabeled dataset.

Feature extraction is a pivotal aspect of both the models. We use BERT to encode contextual meaning from words in reviews by considering the word relationships in a sentence. BERT embeddings enable the model to understand subtle linguistic hints that may hint at fraudulent activity. We also

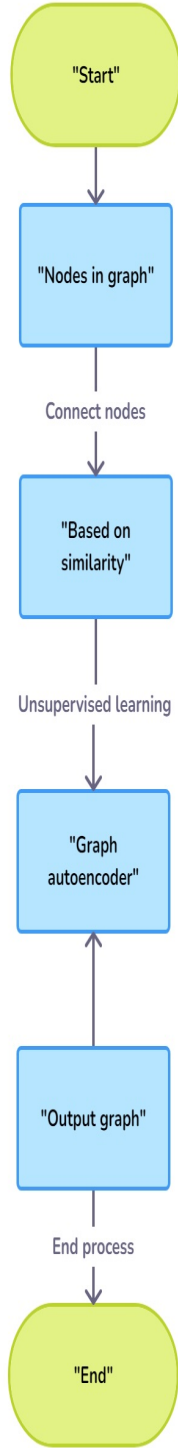


Fig. 1. Graph autoencoder learning latent structures in an unlabeled review dataset

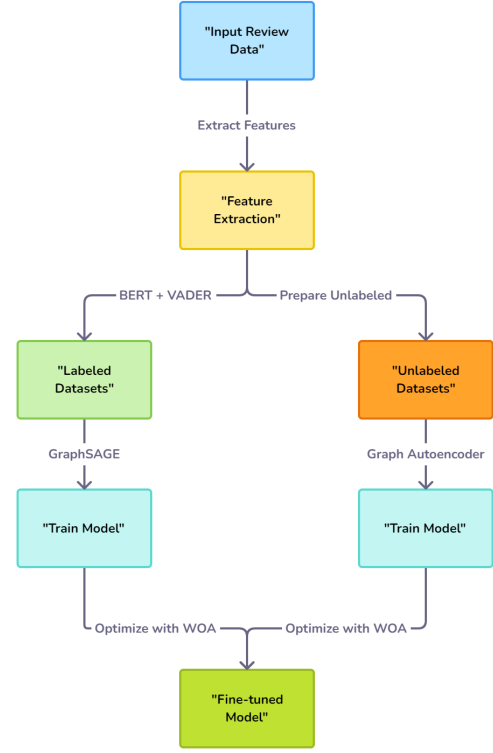


Fig. 2. WOA optimization process for fine-tuning GraphSAGE and graph autoencoder model

employ Valence aware dictionary and sentiment reasoner (VADER) to extract sentiment scores for each review that further distinguish between the forged and authentic versions over their emotional tone. Along with the deep contextual embeddings of BERT and VADER employed for sentiment analysis, a rich feature set is constructed for usage by both GraphSAGE and the graph autoencoder.

To this effect, we incorporate a metaheuristic optimization technique referred to as Whale Optimization Algorithm (WOA), inspired by the hunting mechanism of humpback whales, to fine tune the hyperparameters of GraphSAGE, and graph autoencoder. WOA works efficiently to navigate the high-dimensional search spaces in finding the right set of parameters of a model. Optimizing parameters related to learning rates and size of hidden layers, WOA enables the models to achieve a better trade-off among complexity and performance, resulting in reduced overfitting along with enhanced generalization to new data. Figure 2 presents how WOA is incorporated into our model pipeline and optimized its parameters for better fraud detection.

Finally, our approach includes GraphSAGE for labelled data and graph autoencoders for unlabelled data towards forming a powerful hybrid framework for detecting fraudulent reviews. The hybrid method is further complemented by contextual word representation from BERT and WOA for model optimization, providing a holistic and robust solution to the constantly increasing number of fake reviews in online platforms. By the application of both supervised and unsupervised learning, we can actually detect fraud patterns in various

datasets so as to improve the accuracy and reliability of online review systems. This paper further researches fraud detection: towards a more complex approach in order to be upfront with the dynamic nature of the process through fraudulent reviews.

A. Motivation

The speedy emergence of online review sites has transformed the prevalent trend in consumer behavior. However, fake reviews lead to a further eroding of trust and decision-making capability, making proper classification an immediate need for authenticity determination and improved user experience. This paper introduces a hybrid approach that combines semi-supervised learning with graph based techniques-GraphSAGE, Graph Autoencoders and incorporating the Whale optimisation algorithm. It captures very fine relationships in review data by using labeled and unlabeled data and thus enhances the process of classification. This work is aimed at creating a reliable solution to the challenge of classification of reviews. We found that other versions suffers from several drawbacks; some of them are: the drawback in the feature limitation of existing fraud detection methods such as sentence weight networks and recurrent convolutional neural networks (RCNNs) is that they involve predefined features. Therefore, these methods fail to effectively adapt to the complexity and dynamics of natural language because of predetermined features, which limit their flexibility and accuracy in the detection of fake reviews. For example, even in the case of RCNN, the contextual nuances in reviews tend to escape its perceptivity, especially about fraud tactics. In addition, scalability is the problem when dealing with large amounts of data, which leads to performance degradation of the models under detection.

Some statistics of real scenarios will demonstrate its scale. As of 2021, over 42 percent of online reviews from the major e-commerce marketplaces, such as Amazon, were likely fraudulent or manipulated, giving billions in lost consumer trust and revenues to the businesses. Fake reviews, for instance, cost international consumers approximately 152 billion dollars annually because e-commerce websites are really behind in keeping up with changing ways of committing review fraud. Moving forward with such challenges and the inflexibilities in the earlier models of detection is needed with a design that is much more flexible and scalable as the GraphSAGE and graph autoencoders do support. Both methods are adaptable in terms of the possibility of processing enormous amounts of data the moment new patterns of fraud are detected.

B. Contribution

The contribution of this paper is developing a hybrid approach that classifies online reviews as fake or genuine by integrating semi-supervised learning with graph based techniques. GraphSAGE and Graph autoencoders are considered in the proposed method to improve the model so that it can take full advantage of both labeled and unlabeled data for designing intricate relationships among reviews. Traditional approaches lack this accuracy and robustness. Further, unique graph construction is considered for feature extraction optimization, and

it has enhanced the review's classification capabilities. In all, this paper proposal here provides a comprehensive approach that pushes forward the frontiers of review classification.

C. Evaluation Strategies

We propose a hybrid model for fake review detection: GraphSAGE for labeled data, Graph Autoencoders for unlabeled data, and WOA for hyper parameter optimization. The performance of both labeled and unlabeled data are compared separately by graph-based methods on the former and unsupervised learning on the latter. It applies key metrics, including accuracy, precision, and the AUC, to test the model's efficiency in identifying forged reviews.

This hybrid approach outperforms traditional machine learning methods because it captures relational information as well as latent structures that are inherent within review data. Traditional machine learning models are based on predefined features, which are commonly static and cannot adapt to the changing patterns of fraudulent activities. In contrast to this, the GraphSAGE-Graph Autoencoders-based model analyzes not only the individual review content but also takes advantage of the interactions and similarities between reviews towards accurate classification. Furthermore, this approach is much more computationally efficient than other graph-based models, such as standard GNNs, as GraphSAGE uses a sampling approach to aggregate neighbors, resulting in a massive reduction of computational cost without any sacrifice of performance.

The unlabeled data are clustered based on the embeddings generated by the Graph Autoencoder using unsupervised clustering. Using the WOA for optimization of the hyperparameters shows the best effectiveness of graph-based techniques and the clustering methods, reducing the error and increasing performance. Since WOA optimizes the settings of the model with respect to its balance between complexity and efficiency, accuracy and quality of clustering are expected to be improved when comparing performance both before and after optimization.

A combination of WOA is the proposed hybrid model, which also performs better compared to other traditional methods. The proposed hybrid model further optimizes various significant parameters for learning rate, layer depths, and thresholds of clustering so that it makes the system more efficient, scalable, and potentially capable of processing larger datasets; however, it remains highly accurate with a reduced computation. Finally, this approach presents a scale and adaptively tailored algorithm to detect fraudulent reviews that can outperform standard machine learning techniques and other graph-based models.

D. Paper Organization

The rest of the paper is organized into 6 sections. Section II describes related work. Section III states the Whale Optimization (WOA) model. Section IV explains the methodology of the model. Datasets, and Results and discussion are discussed in Section V and VI respectively. Finally, conclusions are discussed in Section VII

II. RELATED WORKS

The paper [1] discusses the use of Graph neural networks (GNNs) for analyzing graph data, highlighting their effectiveness in capturing complex relations within heterogeneous networks & compares MP-GT with existing models like SA-GCN and CAP. The authors of [2] addressed the significant impact of online consumer reviews on purchasing decisions, highlighting the challenge posed by spurious reviews in e-commerce. In [3], the paper addresses the challenges of edge detection in colour images by UAVs, highlighting issues such as noise and distortion & proposes an improved Whale optimization algorithm (WOA) that utilizes quaternion representation for better edge detection results. The authors of [5] introduced a novel fuzzy optimized convolutional neural network aimed at enhancing the accuracy of user opinion predictions. In [6], the authors highlighted the limitations of existing datasets, which are predominantly in English. Also, previous research has explored various Machine learning (ML) techniques like Naive Bayes and Support Vector Machines, but this paper aims to create a baseline using modern language models like BERT and ELECTRA. Traffic networks are complex and exhibit spatial-temporal dependencies, making accurate prediction challenging and Recent advancements in deep learning have led to various models focusing on spatial-temporal dependency modeling, primarily using convolutions to separate spatial and temporal correlations but real-time traffic state forecasting remains difficult due to the intricate nature of traffic networks.

So, review [7] proposes a hybrid deep learning framework for traffic prediction which can capture dynamic dependencies simultaneously. The authors of [8] found out the problem in traditional methods for node embeddings - graphs require all nodes to be present during training, which limits their ability to generalize to unseen nodes. So, they introduced GraphSAGE, an inductive framework that generates embeddings for unseen nodes by using local neighborhood features of the node enhancing scalability and adaptability in dynamic graph scenarios. In literature [9], the authors gave the result that the whale bionic algorithm reduced prediction error and gave greater accuracy when compared to the prediction results of models-RNN, LSTM, WOA-LSTM. In [10], the authors addressed problem of increase in nonlinear loads in power systems which led to significant harmonic distortion in voltage and current, necessitating effective solutions. So, the authors used Whale optimization algorithm (WOA) to employ Selective Harmonic Elimination method to optimize inverter output voltage waveforms and eliminate low-order harmonics more efficiently.

The paper [11] addresses the challenges of managing grid stability due to the variability of renewable energy sources like wind and solar, which can cause transient overvoltage and other disturbances. Hence, The whale optimization algorithm is proposed as a solution optimizing the allocation of resources and control parameters and improving frequency regulation. The paper [12] discusses Author Gender Detection as a critical issue in communication under internet security. So, research employs ML and meta-heuristic algorithms, specifically an

artificial neural network (ANN) combined with the Whale optimization algorithm to improve classification accuracy. In [13], the authors aim to enhance fraud detection by employing a hybrid approach that combines supervised and unsupervised algorithms, specifically Light Gradient Boosting Machine (LGBM) and Kernel Principal Component Analysis (KPCA) to improve the Receiver Operating Characteristic-Area Under Curve.

Credit card fraud is a significant issue in digital transactions, highlighting the need for effective detection methods as traditional methods often fail to capture complex fraudulent patterns. The study [14] focuses on outlier detection as a promising method to identify fraudulent activities within transactional data. The paper [15] addresses the challenge of understanding disease associations through biological data, emphasizing the need for effective models to analyze complex genetic interactions. GraphSAGE - a method to generate embedding for nodes in a graph, for scalable learning on large graphs by sampling and aggregating features from a node's neighbors and GCN is used to perform convolution operations on graph data, to capture the relationships between nodes and their features. In study [16], the authors have introduced a method Traditional Image Steganography for embedding messages in the images so that the messages will not be detected by steganalysis techniques. The development of GANs has enabled the implementation of this method enhancing the effectiveness of steganography and effectively resisting both CS-steganalysis and RF-steganalysis. The paper [17] investigates the challenges of AI-based fraud detection system, focusing on real-time transaction surveillance and anomaly detection to enhance digital payment security. The study employs a comparative exploration approach such as integration of AI technologies, such as machine learning and data analysis, to continuously monitor transactions and identify fraudulent activities.

In [18], the authors introduced the Intelligent Learning Scheme for Digital Fraud Detection, a deep learning-based engine that utilizes feature selection and classification principles to detect credit card fraud. The paper [19] employs a systematic process to gather and analyze research on spam review detection, focusing on recent trends and characteristics of data, features, and models used in various methods. In [20], The authors discussed the influence of online customer reviews and the rise of fake reviews & highlighted the use of ML techniques for detecting fake reviews & emphasizing the importance of feature extraction methods like TF-IDF.

The authors of [21] suggest a need for integrating psychological insights and computational linguistics to enhance detection methods. Additionally, a hybrid review approach can help identify research gaps and future avenues. The development of the internet and online platforms such as social media feeds, news blogs, online newspapers have led to manipulation of news shaping public opinion in a matter of time. So, the authors of [22] proposed a hybrid system to detect fake news articles based on content-based features and the WOA-XGBTree algorithm optimized by the Whale Optimization Algorithm to classify news articles using extracted features. The increasing prevalence of online review systems has led

to the rise of fake reviews, where malicious actors such as competitors and service providers manipulate consumers by creating deceptive reviews using Sybil accounts, bot farms, or authentic accounts. So, the review [23] proposes a novel approach for detecting fake reviews using Logistic Regression, a machine learning technique. By applying this method, the model can predict the likelihood of a review being fake based on specific features, improving the detection process.

In [24], GNN HHO outperformed other models with 87.6 F-1 score and BERT features enhanced model performance significantly. Also, Metaheuristic

optimizations improved GNN model performance. In [25], the authors present an ensemble ML model to identify whether a review is fraudulent or authentic. To achieve this objective, amazon reviews dataset is used & the proposed model outperformed as compared to others as Random Forest algorithm provides 99% accuracy which is better than other algorithms. By proposing a fuzzy automatic disturbance rejection control (ADRC) method enhanced by an Improved whale optimization algorithm (IWOA), the authors introduce logistic chaos mapping and skew tent mapping to enhance population diversity and global search capabilities. The authors of [26] propose a channel biattention CNN module to adaptively emphasize the importance of relevant prior knowledge to enhance the model's ability to accurately classify Consumer fraud which involves illegal activities aimed at generating revenue.

In [31], paper discusses the challenges of solving the first Fredholm integral problem related to particle size distribution highlighting the limitations of conventional optimization algorithms, whale optimization algorithm (WOA) is presented as a promising alternative due to its global performance, speed, and ability making it suitable for complex particle size inversion problems. In [32], the proposed SAGE-Net addresses the limitations of existing methods by integrating semantic and geometric features, enhancing the robustness of global descriptors for place recognition. Experiments on benchmark datasets indicate that SAGE-Net outperforms state-of-the-art approaches, demonstrating its effectiveness and ability in unseen scenes. The authors discuss the automated synthesis of analog circuits, highlighting various frameworks and methodologies emphasizing a hierarchical design approach where circuit blocks are built from fundamental analog primitives for creating complex system-level designs [33]. GNN generalizes CNN to graphs, inspired by Fourier transformation & Heterogeneous GNN methods transform graphs, apply GNN, and aggregate representations in [34].

The model presented in [35], which denoted a new graph neural network based for feature extraction from review networks to enrich fake review detection, was implemented with GEM. The authors also suggest a more precise enhancement to GNNs and a multi-feature detection framework, to better detect fraudulent behaviors. In [36], they propose a fine-grained deceptive opinion spammer detection scheme for fraudulent review reviews, and therefore present the first graph neural network framework to address the inadequacy of spoofing based verification procedures in online opinions by stressing user interactions and conducting social context with an emphasis on customer service quality while detecting

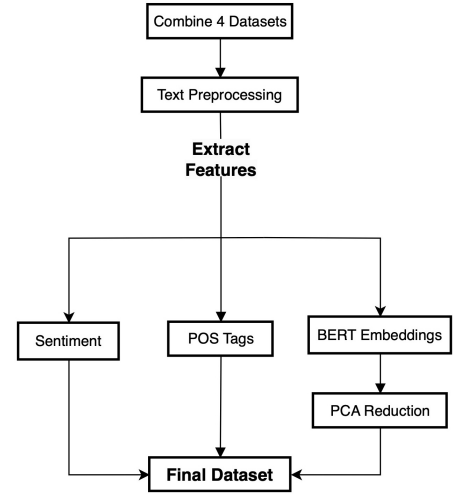


Fig. 3. Proposed feature selection approach

fake reviewers. Given the comprehensive survey done on fake reviewer detection by [37] authors, they classified the existing fake review detector methods among 3 major streams as Text-based, Behavior-based and Graph based approach which also mention about evolution and challenges in field of fake reviewer detection stating where we have reached till June 2019 to introduce our proposed RHGNN framework.

The paper [39] introduces GraphSAGE, a graph-based approach, enhanced with a covertness model to improve the detection of fake reviews by analyzing user interactions and review content. The proposed model shows competitive performance in Recall and AUC metrics when tested on real-world Amazon datasets, outperforming other models like GCN and GAT. Researchers of [40] combined the whale optimization algorithm (WOA) for feature selection with the rotation forest algorithm to classify emails as spam or non-spam. Remarkably, this hybrid method achieved a 99.9% accuracy in spam classification, significantly outperforming previous techniques.

III. WHALE OPTIMIZATION ALGORITHM (WOA) MODEL

The Whale optimization algorithm was used to optimize hyperparameters of the GraphSAGE and Graph autoencoder model for fake review detection in this project. WOA is based on the mechanism ang hunting behavior of humpback whales. WOA was used to optimize the learning rate, which determines how quickly the model learns.

$$X(t+1) = X(t) - A \cdot D \quad (1)$$

The WOA updates parameter values using a position update formula, where $X(t)$ is the current best solution, A - coefficient vector controlling exploration, and D - distance to the best solution.

The WOA consists of three stages - encircling prey (candidate solutions move closer to the optimal), bubble-net foraging (search space is shrunk as if a spiral around the prey), and search for prey, in which exploration is expanded. In the classification task, for 50K reviews (2 labeled, 2 unlabeled datasets) WOA optimized GraphSAGE (for labeled data) and Graph

TABLE I
PERFORMANCE COMPARISON TABLE OF SUPERVISED LEARNING METHODS

Methods	Metrics	Performance (%)	Dataset	Reference
SVM	F-measure	78.1	Yelp, etc	[43]
LIBSVM	Accuracy	89.6	Yelp, etc	[44]
NB-SVM	Accuracy	91.9	IMDB	[45]
WMUSVM	Recall	82.5	Trip Advisor	[46]
BERT	Accuracy	90.5	Yelp	[47]
XGBoost	Precision	99.0	Yelp	[48]

TABLE II
COMPARATIVE ANALYSIS TABLE

Year	Ref.	Approach	Classifier	Dataset	Acc.
2024	[24]	Metaheuristic Algorithm	1. Random Forest, 2. Feed-forward NN, 3. KNN, 4. Logistic regression	Amazon reviews	75.92%, 75.92%, 78.87%, 84.07%
2021	[38]	Improved Whale Optimization	1. MLP, 2. CS, 3. GWO, 4. BA	1. FNC-1 dataset, 2. ARC dataset, 3. Claim polarity dataset, 4. Perspectrum dataset, 5. Snopes dataset	76.53%, 74.72%, 78.45%, 79.63%, 59.02%
Proposed model	-	Metaheuristic Algorithm	1. GraphSAGE, 2. Graph AutoEncoder	1. Fake Reviews (40K), 2. Yelp Labeled dataset, 3. iPhone Reviews, 4. Google Maps Restaurant	98%

Autoencoders (for unlabeled) could better capture relational and textual information of the dataset which resulted a higher classification accuracy.

This process helped GraphSAGE, Graph autoencoder model capture the relations between reviews by doing an effective tuning of the graph structure reflecting relational and textual information. Finding that optimal balance of the hyperparameters algorithm efficiently enhanced model's capability to classify between realistic and fake reviews.

IV. METHODOLOGY

This project uses a hybrid approach toward classifying about 50K reviews from Amazon, Yelp, iPhone, and Google Maps as true or spam. Following all preprocessings-stemming, removal of stopwords, and lemmatization, features are extracted using BERT embeddings for semantic context and VADER for sentiment analysis. GraphSAGE is used for generating graph-based embeddings for the labeled data (50K reviews) by aggregating information from neighboring nodes that, in turn, support the supervised learning process. Graph Autoencoders discover anomaly patterns in the reviews by using unsupervised learning. Lastly, the hyperparameters are adjusted with the Whale optimization algorithm to allow flexibility in the ability to improve the model. Finally, results of classification of the review as real or fake are evaluated through probability score, along with measures of accuracy, precision, recall, and F1-score.

A. Preprocessing

The concatenated dataset consists of approximately 50,000 reviews from four datasets - product reviews on E-commerce platforms (Amazon and Yelp), iPhone reviews, and Google Maps restaurants reviews.

Preprocessing starts with the stemming and lemmatization process, this process reduces words into their root and base form. For example, word-running is reduced to 'run'. Secondly, Non-text handling, star ratings ranging from 1 to 5 are normalized, Normalization scales numerical features to a standard range. Thirdly, Stopword removal words that do not add meaning to the review, words like 'the', 'is', and 'are' removed from the text data, these words do not contribute to distinguishing between computer-generated and genuine reviews.

B. Feature extraction

We implemented feature extraction pipeline that analyzes preprocessed reviews from multiple perspectives — capture emotional tone, linguistic structure and semantic meaning, started with loading pre-processed data.

VADER: We applied Valence Aware Dictionary and sEntiment Reasoner (VADER) to extract four sentiment scores for each review. Positive score reflected the extent of positivity in the review, Negative score to measure how negative the sentiment in text is, Neutral score measured the presence of neutral language in a review, Compound score combined all three scores to create an overall sentiment metric that summarized both the emotional polarity and strength of feelings conveyed by review. With the combination of these four scores we were able to measure better emotional tone in reviews.

Part-of-Speech (POS) tagging: We used part of speech (POS) tagging to measure linguistic structure, to identify and count specific word types - nouns, verbs and adjectives, occur in each review. We counted the number of nouns indicating subjects in text, verbs representing action-filled language using Natural language toolkit's POS tagging function and adjective count as a proxy for describing intensity. We used these POS

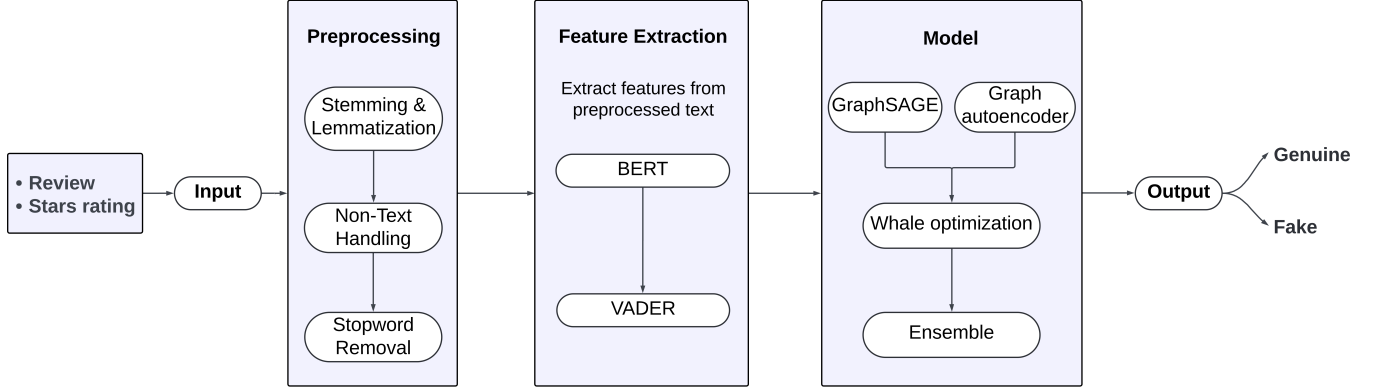


Fig. 4. Flow diagram of proposed hybrid model

counts to measure the linguistic depth of each review and better understand what style aspects were being covered by a reviewer.

$$\text{POS count} = \sum_{i=1}^n \delta(\text{POS}(\text{word}_i), \text{target POS}) \quad (2)$$

where n is total number of words in the review, $\text{POS}(\text{word}_i)$ represents the POS tag of the i -th word, and δ is the Kronecker delta function.

BERT: We utilized Bidirectional Encoder Representations from Transformer model to understand the semantic content. This model was used to generate embeddings (high-dimensional vector representations of each review). But these high dimensionality possessed challenges for computational efficiency. To address this, we applied Principal component analysis (PCA) to reduce each embedding to three principal components. Finally, we combined all the features that we extracted into one dataset. This included VADER sentiment scores, POS counts, review length, and PCA-reduced BERT embeddings. Raw sentiment score R is calculated as the sum of the sentiment scores for each token in the text:

$$R = \sum_{i=1}^n \text{score}(\text{token}_i) \quad (3)$$

where n represents the total number of tokens and $\text{score}(\text{token}_i)$ is the sentiment score assigned to each token.

Compound sentiment score C is a normalized version of the raw score, scaled to a range between -1 and +1:

$$C = \frac{R}{\sqrt{R^2 + \alpha}} \quad (4)$$

where R is raw score, and α is constant which controls normalization factor. By combining these different features, we got a multi-dimensional view of each review encapsulating both surface-level sentiment and deep semantic meaning. The combined dataset was then saved as a CSV file.

C. Graph models

1) *GraphSAGE for labelled data:* In our fraud review detection system, GraphSAGE is chosen as the primary algorithm for supervised learning over labeled datasets, consisting of approximately 50,000 reviews. It is because GraphSAGE has efficiency in learning from large-scale graph data; besides, node attributes and local neighborhood structures are really important in many scenarios to uncover hidden patterns. This makes it a strong technique for review fraud detection; relationships between reviews are able to discover suspicious patterns that the simple feature-based approach may not discover.

The first preferred method for review classification is GraphSAGE Graph Sample and Aggregate because of its inductive capacity. Unlike transductive methods such as the classical GCNs, which would require retraining in case more data is added, GraphSAGE can generalize on unseen nodes during inference. This inductive property proves to be very useful for fraud detection purposes on dynamic e-commerce platforms where new reviews are being generated all the time. It lets us use GraphSAGE in real-time fraud detection without requiring us to retrain the model from scratch whenever new reviews come up. GraphSAGE aggregates information from a node's local neighborhood, thereby capturing both local and global structural patterns in the graph. This is very important in the context of our review classification task because fraudulent reviews tend to reflect unique relational structures compared to the original ones. The use of GraphSAGE allows the network to detect a pattern of collusion or concerted effort to inflate or deflate the products in question. This relational dimension is crucial for fraud detection because it would raise the chances of capturing anomalies that would result from the interaction rather than from the review content itself.

GraphSAGE is good at review classification by virtue of its node embeddings generation. Each review here is a node, edges representing content similarity through the cosine similarity of its BERT Embeddings. GraphSAGE aggregates the information that it has learned from the neighbors and

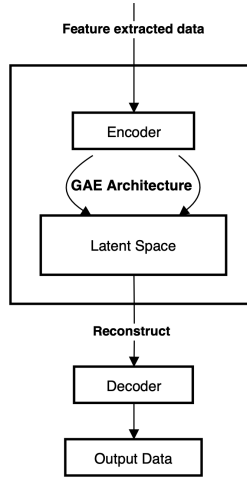


Fig. 5. Architecture of the graph autoencoder model for unlabelled datasets

captures localized patterns. For node classification, we do the following graph structure: $G=(V,E)$, where V denotes the set of nodes (reviews) and E is the set of edges (similarity connections). The algorithm samples neighbors for each node, aggregates their features, generates node embeddings, and uses these for supervised learning with the cross-entropy loss. This makes GraphSAGE an ideal model to capture review content along with local structural information, thus suitable for fraud detection within labelled reviews. This gives it an edge in suitability for dynamic datasets, like reviews, because it can generalise to unseen nodes.

2) *Graph autoencoder for unlabelled data:* A graph-based model was used in order to enhance the representations both on the labeled and unlabeled examples. The data-set of concatenated vectors from the labeled and the unlabeled samples was used in the building the KNN graph. Then, the adjacency matrix produced by the KNN graph was converted into an edge index format which is more suitable to be used with PyTorch Geometric's data structure. Such a representation was fed to a Graph autoencoder (GAE) for training to provide low-dimensional embeddings from the node-level features. Equation 5 of the decoder in graph autoencoder. The decoder in the graph autoencoder reconstructs the adjacency matrix $\hat{A}$ from the latent embeddings Z . This can be expressed mathematically as follows:

$$\hat{A} = \sigma(ZZ^T) \quad (5)$$

As shown in Equation 5, the reconstructed adjacency matrix $\hat{A}$ is obtained by applying the activation function σ to the product of the latent embedding matrix Z and its transpose Z^T .

The two layers of GCN were used in this GAE model architecture. Input feature space is first transformed into a hidden dimension of 16 by the first GCN, and then by the second GCN into 8 dimensions. ReLU activation was applied between layers to introduce non-linearity. At train time, the reconstruction of graph structure was optimized through an unsupervised loss function computed over edge connections. We apply early stopping for the purpose of avoiding overfit-

Algorithm 1 Graph autoencoder training

Require: Data X , Neighbors $k = 5$, Epochs $T = 20$

Ensure: Node Embeddings Z

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1:  $G \leftarrow \text{KNN}(X, k)$   $\triangleright$  Create a graph from data using
   k-nearest neighbors
2:  $V \leftarrow \text{Nodes from } G$ 
3:  $E \leftarrow \text{Edges from } G$ 
4: Initialize GAE model: Encoder + Decoder
5:  $\text{Optimizer} \leftarrow \text{Adam optimizer}$ 
6: for  $epoch = 1$  to  $T$  do
7:    $Z \leftarrow \text{Model.encode}(V, E)$   $\triangleright$  Encode node
   embeddings
8:    $\hat{E} \leftarrow \text{Model.decode}(Z)$   $\triangleright$  Reconstruct graph edges
9:    $Loss \leftarrow \text{Reconstruction loss between } E \text{ and } \hat{E}$ 
10:   $\text{Optimizer.zero_grad}()$ 
11:   $Loss.backward()$   $\triangleright$  Backpropagation
12:   $\text{Optimizer.step}()$   $\triangleright$  Update model parameters
13: end for
14: return  $Z$   $\triangleright$  Return final node embeddings

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ting; after that, we extract learned embeddings for both labeled and unlabeled data once it has converged so that we can further use them for classification tasks.

D. Metaheuristic optimization

In this paper, we try to narrow our research questions in a way that the experimental approach would be guided by clear objectives fitting into the larger purpose of improving fraudulent review detection. Our first research question is: 'How does GraphSAGE combined with a graph autoencoder improve classification accuracy compared to traditional methods for detection?' Based on this hypothesis, the question being formed is whether using a hybrid model would outperform standard machine learning techniques by utilizing the labeled data for GraphSAGE and unlabeled data on a graph autoencoder. As it captures relational as well as latent structures within the data, we hypothesize that the model would greatly enhance accuracy in the detection of fraudulent reviews.

E. Output generation

Results of hyperparameter optimization for the LightGBM classifier using WOA is shown in the next section. Optimization was run with configuration of ten whales and three iterations of optimization of the following hyperparameters: the number of leaves, maximum depth, and the learning rate, specified within designed bounds for feasible values. The WOA itself succeeded in finding the following optimal hyperparameters: the number of leaves = 42, max depth = 10, learning rate = 0.15.

These parameters were then used to create the LightGBM classifier and trained on the 40,000 labeled dataset with results classifying review as genuine or fake.

Algorithm 2 Whale optimization algorithm for Hyperparameter tuning

Require: Data X , Labels y , Number of whales n , Max iterations T , Hyperparameter Bounds $[minx, maxx]$

Ensure: Best hyperparameters X^*

```

1: Initialize  $n$  whales with random positions in range  $[minx, maxx]$ 
2: Calculate fitness of each whale using a scoring function
3: Set best position  $X^*$  and best score  $F^*$  found
4: for each iteration from 1 to  $T$  do
5:   for each whale do
6:     Calculate distance to the best position
7:     Update whale's position based on this distance
8:     Ensure position stays within  $[minx, maxx]$ 
9:     Calculate the fitness of the new position
10:    if this fitness is better than  $F^*$  then
11:      Update  $X^*$  and  $F^*$  with new values
12:    end if
13:  end for
14:  if no improvement in best score for 5 iterations then
15:    stop ▷ Terminate the process
16:  end if
17: end for
18: return  $X^*$  ▷ Return the best hyperparameters found

```

V. DATASETS

A. Fake Reviews dataset

The total dataset [50] was 40,000 product reviews with an equal number of both authentic and fake reviews, which amounts to 20,000 for each group.

B. Yelp labeled dataset: Spam reviews for New York

This dataset [51] consists of 20,000 labeled hotel reviews for New York City hotels from Yelp, down selected from an initial set of 350,000 reviews. Each review is labeled as genuine or fake, and reviews labeled 1 are actual reviews, whereas -1 labeled reviews are spam or fake reviews.

C. iPhone reviews from Amazon.com

This is an unlabelled dataset [52]. It contains over 5K reviews of Apple iPhones from Amazon platform. Customer ratings, helpfulness votes, and review text are provided.

D. Google maps restaurant reviews

This unlabeled dataset [53] consists of about 1K reviews of different restaurants from Google Maps.

VI. RESULTS AND DISCUSSION

This section presents the results obtained from the proposed model for the classification of fake reviews. The classification part was implemented with the embedding generation process using GraphSAGE, Graph autoencoder, optimized by the Whale Optimization Algorithm, with LightGBM classifier. This hybrid approach manages to make an effective classification of reviews as well as both textual as well as structural

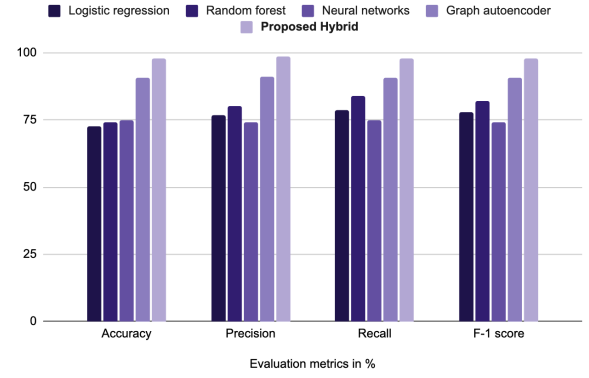


Fig. 6. Comparison of evaluation metrics of proposed hybrid model with state-of-the-art models on 40,000 dataset (Proposed model with highest evaluations)

features. For the evaluation of the effectiveness of the model in the differentiation between genuine and fake reviews, the proposed performance metrics include accuracy, precision, recall, and F1-score.

The hybrid model was tested with Whale optimization, LightGBM classifier on a combined dataset of 40,000 reviews.

A. Model Performance Overview

Overall dramatic improvements in accuracy and performance were seen with the WOA optimized version. Not being optimal initially without the WOA, classification proved to be less effective by a considerable margin. Yet, with WOA introduced to fine-tune hyperparameters of LightGBM, the model turned highly precise towards representing complex patterns present within the data. Therefore, the improvement points to WOA's influence over whether the model could differentiate the real review from a fabricated one. With the hybrid approach that involves GraphSAGE embeddings and optimal hyperparameters, the model gave reliable classification results, converged within a balanced period of training, and therefore reinforced its applicability in real-world application.

B. Confusion Matrix Analysis

Deep exploration of model performance is presented using the confusion matrix Figure 7, which describes true positives and negatives, false positives and negatives. The model classifies a count as high as 27,880 valid reviews and 12,120 spurious reviews, with very low counts in the misclassifications category. Minimal counts in the false positives and false negatives categories help preserve user confidence by implying that the model rarely classifies spurious reviews as valid.

C. Precision Recall Curve Analysis

The Precision-Recall curve (Figure 8) displays precision-recall trade-offs of the model, which is very important for imbalanced data sets like this. A significant area under the curve shows the model has a high precision but remains consistent with that, which helps to significantly decrease false positives while detecting fake reviews.

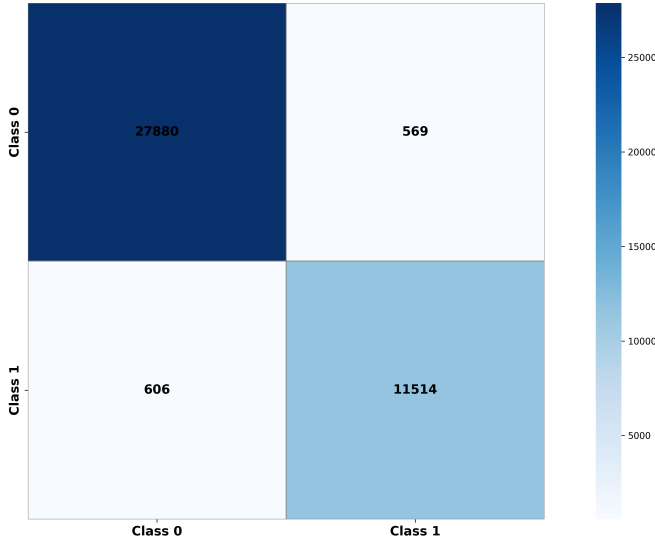


Fig. 7. Confusion matrix of proposed hybrid model, where Class 0 refers to genuine reviews and Class 1 as fake

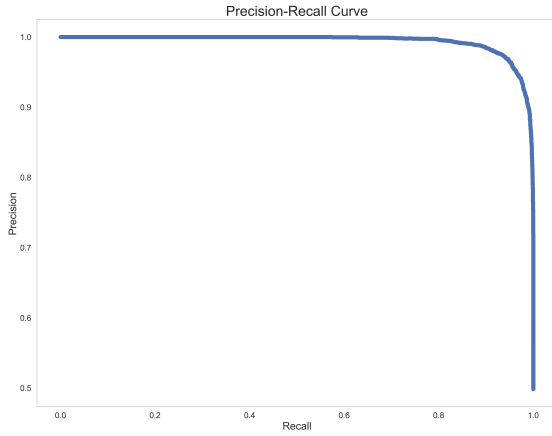


Fig. 8. Precision-Recall Curve of Model Performance

D. 3D Embedding Visualization

As indicated in (Figure 9), the GraphSAGE model produces embeddings that are represented in a 3D scatter plot, effectively communicating the data in reduced dimensions. Each point in this plot is a review; it is encoded as a vector to capture the learned features of the model. By plotting along three axes of embeddings (Embedding 1, embedding 2, and embedding 3), we could clearly notice clusters where the real reviews lie in one cluster and the fake reviews in another. This clustering means that the model is successful in making a clear distinction between real and fake reviews depending on the learned representations of the same. Hence, this 3D plot depicts that the hybrid model is successful in organizing and separating the data based on key features pointing towards the authenticity of a review.

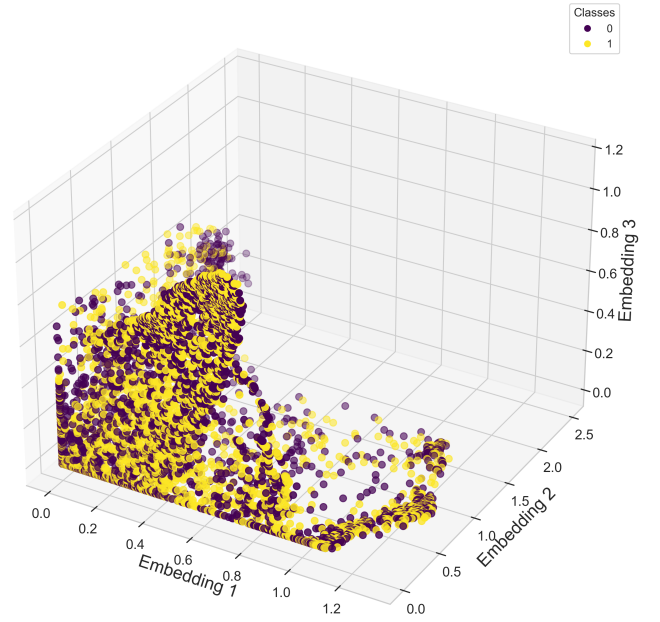


Fig. 9. 3D visualization of learned embeddings (Embedding 1, 2, and 3) capturing complex patterns distinguishing real from fake reviews, based on relational and sentiment-driven features

TABLE III
CLASSIFICATION REPORT OF THE PROPOSED HYBRID MODEL

Class	Precision	Recall	F1-Score	Support
Genuine (0)	0.98	1.00	0.99	27,880
Fake (1)	0.99	0.95	0.97	12,120
Accuracy	-	-	0.98	40,000
Macro Avg	0.99	0.98	0.98	40,000
Weighted Avg	0.98	0.98	0.98	40,000

E. Interpretation of Results

This hybrid model does work well in accuracy, precision, and recall for the classification of fake reviews, which proves it to be robust. GraphSAGE embedding has been leveraged well with LightGBM to make the best use of graph structure and detect fake reviews. Further, the hyperparameters obtained optimized via WOA have served to boost performance: as it shows fewer mis-classifications.

The evaluation metrics used in classification models are defined as follows:

Accuracy - measures overall correctness of the model by calculating ratio of correctly predicted instances (fake and genuine reviews) to the total instances in the dataset:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (6)$$

where TP = True positives, TN = True negatives, FP = False positives, and FN = False negatives.

Not only it is accurate for classification at high levels, but also this method has useful training times, so it is relevant for real-world applications. The patterns from the clustering in Figure 6 highlight the effectiveness of graph-based embeddings, thus supporting our hypothesis that structural data does enhance the classification accuracy on fake review detection tasks.

VII. CONCLUSION

In conclusion, fake reviews remain a significant challenge in the e-commerce industry. In this work, we wrote a graph-based fraud detection methodology that combines GraphSAGE for labeled data and Graph autoencoders for unlabeled data with advanced feature extraction techniques to demonstrate how structural and semantic features can be used to identify fraudulent review behavior. Indeed, this model performs well illustrated by F1 score of 98% outperformed other popular classification algorithms. Further validation of the strength of this model could be explored by extending it to other types of user-generated content, such as social media or blog comments. Another exciting field to be explored is the ability of real-time fraud detection, where online platforms are able process data instantly in order for review policies that prevents fake reviews as they come up. Future research, beyond model optimization also needs to focus on the development of heuristics specific for domains, rules based systems or hybrid models that combine traditional methods alongside graph-based approaches in order to scale down without depending overwhelmingly on AI.

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